

Wi-Fi Training Program

Module 5 – Assignment Questions & Answers

Sri Ragavendra S

1. What are the key features of Wi-Fi 6, 6E and 7 and how do they differ from previous standards like Wi-Fi 5 (802.11ac)?

Answer:

Wi-Fi 6 (802.11ax), Wi-Fi 6E, and Wi-Fi 7 (802.11be) represent a significant leap from Wi-Fi 5 (802.11ac), shifting the focus from peak speed for a single device to efficiency, capacity, and low latency for many devices simultaneously.

Feature	Wi-Fi 5 (802.11ac)	Wi-Fi 6 (802.11ax)	Wi-Fi 6E	Wi-Fi 7 (802.11be)
Standard	802.11ac	802.11ax	802.11ax (extended)	802.11be
Frequency Bands	5 GHz only	2.4 & 5 GHz	2.4, 5 & 6 GHz	2.4, 5 & 6 GHz
Max Throughput	3.5 Gbps	9.6 Gbps	9.6 Gbps	46 Gbps
Channel Width	Up to 160 MHz	Up to 160 MHz	Up to 160 MHz	Up to 320 MHz
Modulation	256-QAM	1024-QAM	1024-QAM	4096-QAM
Multi-User	DL MU-MIMO (4 users)	UL+DL MU-MIMO (8 users)	UL+DL MU-MIMO (8 users)	MU-MIMO (16 users)
Channel Access	OFDM	OFDMA	OFDMA	OFDMA + MLO
Spatial Streams	8	8	8	16
Power Save	Basic PS	TWT (Target Wake Time)	TWT	Enhanced TWT
Key Innovation	MU-MIMO, 160 MHz	OFDMA, TWT, BSS Color	6 GHz clean spectrum	MLO, 320 MHz, 4096-QAM

Key Improvements Over Wi-Fi 5:

- OFDMA (Wi-Fi 6): Wi-Fi 5 used OFDM where one user occupies the entire channel per transmission. Wi-Fi 6 introduced OFDMA, allowing multiple users to share the channel simultaneously via Resource Units — dramatically improving efficiency in dense environments.
- BSS Coloring (Wi-Fi 6): Reduces co-channel interference by tagging frames with a BSS color. Devices can ignore transmissions from overlapping BSSs with a different color, reducing unnecessary deferrals.
- TWT (Wi-Fi 6): Allows devices to schedule wake-up times, extending battery life — especially useful for IoT.
- 6 GHz Band (Wi-Fi 6E): Opens up 1200 MHz of fresh, uncongested spectrum with no legacy device interference.
- MLO (Wi-Fi 7): Devices can transmit and receive simultaneously across multiple bands/channels, reducing latency and increasing reliability.
- 4096-QAM (Wi-Fi 7): Encodes 12 bits per symbol vs 10 bits (1024-QAM), boosting peak rates by ~20%.
- 320 MHz Channels (Wi-Fi 7): Doubles the channel width available in Wi-Fi 6/6E.

2. Explain the role of OFDMA in Wi-Fi 6 and how it improves network efficiency.

Answer:

What is OFDMA?

OFDMA (Orthogonal Frequency Division Multiple Access) is an extension of OFDM that allows the wireless channel to be divided into smaller frequency allocations called Resource Units (RUs), which can be assigned to different users simultaneously within the same transmission.

How OFDMA Works in Wi-Fi 6:

- In Wi-Fi 5 (OFDM), when the AP transmits, the entire 80 MHz or 160 MHz channel is used for one client at a time. Other clients must wait.
- In Wi-Fi 6 (OFDMA), the AP divides the channel into RUs of different sizes: 26-tone, 52-tone, 106-tone, 242-tone, 484-tone, and 996-tone RUs.
- Multiple clients are assigned different RUs and transmit/receive simultaneously in the same OFDMA transmission.
- The AP uses a Trigger Frame (for uplink) to schedule which clients transmit on which RUs and at what time.
- For downlink, the AP assigns RUs in the HE-MU PPDU using the HE-SIG-B field to signal each client's allocation.

How OFDMA Improves Network Efficiency:

- Parallel Transmissions: Instead of sequential per-client transmissions, multiple clients are served simultaneously — reducing total airtime usage.
- Efficient for Small Packets: Wi-Fi 5 wasted bandwidth when a client had a small frame (e.g., an ACK or DNS query) but still occupied the full channel. OFDMA can assign a small RU (26 tones) for small frames, freeing larger RUs for high-throughput clients.
- Reduced Latency: Clients no longer wait in line — they can be scheduled to transmit simultaneously, cutting queuing delays.
- Better Fairness: The AP scheduler can balance RU assignments based on client needs and priorities.
- Improved Capacity: An AP can serve many more clients per unit time compared to OFDM-based standards.
- Uplink OFDMA: Wi-Fi 6 introduced uplink OFDMA (UL-OFDMA) — multiple clients can upload simultaneously triggered by a single Trigger Frame, which Wi-Fi 5 could not do.

OFDMA vs OFDM Summary:

Aspect	OFDM (Wi-Fi 5)	OFDMA (Wi-Fi 6)
Channel per Transmission	One user gets entire channel	Multiple users share channel via RUs
Small Packet Efficiency	Wasteful — full channel used	Efficient — small RU assigned
Latency	High in dense environments	Low — parallel scheduling
Uplink Multi-User	Not supported	Supported via Trigger Frames
Best For	Single user, high throughput	Many users, mixed traffic

3. Discuss the benefits of Target Wake Time (TWT) in Wi-Fi 6 for IoT devices.

Answer:

What is TWT?

Target Wake Time (TWT) is a power management mechanism introduced in 802.11ax (Wi-Fi 6) where an AP and a client negotiate a specific schedule — defining exactly when the client will wake up to transmit or receive data. Between TWT sessions, the client's radio can remain completely off.

How TWT Works:

- The AP and client exchange TWT Setup frames to agree on: TWT Wake Interval (how often the device wakes), TWT Wake Duration (how long it stays awake), and the TWT Start Time (when the first session begins).
- Individual TWT: A private schedule between AP and one client — ideal for devices with unique data patterns.
- Broadcast TWT: The AP schedules a group of devices to wake simultaneously — used for IoT devices with similar polling intervals.
- During the agreed sleep period, the AP buffers any frames destined for the sleeping client.

Benefits of TWT for IoT Devices:

- Dramatic Battery Life Extension: IoT devices like sensors, smart meters, and wearables transmit small amounts of data infrequently. TWT allows them to sleep for seconds, minutes, or even hours between wake cycles. This can extend battery life by up to 7× compared to legacy power save modes.
- Predictable Wake Schedule: Unlike legacy PS-Poll where devices wake every beacon interval (102.4 ms), TWT lets IoT devices wake only when they actually need to — perfectly matched to their data transmission pattern.
- Reduced Channel Contention: When IoT devices are asleep, they are not competing for the channel during their off periods, reducing congestion for other devices.
- Scalability: An AP can manage hundreds of IoT devices with different TWT schedules without them all competing simultaneously.
- Improved Network Efficiency: The AP knows exactly when each device will be active, allowing it to plan transmissions efficiently and reduce wasted beacon responses.
- Broadcast TWT for Synchronized Polling: The AP can wake groups of IoT devices simultaneously for batch data collection — useful in industrial IoT and smart building applications.
- Reduced Interference: Fewer active devices at any given time means less interference and a cleaner radio environment.

Real-world Example: A temperature sensor that reports every 5 minutes can negotiate a TWT wake interval of 5 minutes, sleeping entirely between reports. With legacy Wi-Fi, it would have to wake every 102.4 ms to check the beacon — consuming ~3000× more wake cycles per hour.

4. Explain the significance of the 6 GHz frequency band in Wi-Fi 6E.

Answer:

What is Wi-Fi 6E?

Wi-Fi 6E is an extension of Wi-Fi 6 (802.11ax) that adds support for the 6 GHz frequency band (5.925–7.125 GHz), in addition to the existing 2.4 GHz and 5 GHz bands. The '6E' stands for '6 GHz Extended'.

Significance of the 6 GHz Band:

- 1200 MHz of New Spectrum: The 6 GHz band provides up to 1200 MHz of additional spectrum — far more than the 70 MHz available in 2.4 GHz or 500 MHz in 5 GHz. This is the largest spectrum addition to Wi-Fi since its inception.
- 7 Non-Overlapping 160 MHz Channels: The 6 GHz band can accommodate 7 completely non-overlapping 160 MHz channels. In comparison, 5 GHz offers only 2 non-overlapping 160 MHz channels, and 2.4 GHz only has 3 non-overlapping 20 MHz channels.
- Zero Legacy Device Interference: Only Wi-Fi 6E-capable devices (those supporting 802.11ax in 6 GHz) can operate in this band. Older 802.11a/b/g/n/ac devices cannot access 6 GHz, so there is no legacy device pollution causing slow beacon responses or protection mechanisms.
- Clean, Uncongested Spectrum: The 2.4 GHz and 5 GHz bands are heavily congested from years of device accumulation. The 6 GHz band starts fresh — no neighboring Wi-Fi networks, no Bluetooth interference (at 2.4 GHz), and no microwave interference.

- **No DFS Requirement:** Unlike parts of the 5 GHz band, the 6 GHz band in most regions does not require Dynamic Frequency Selection (DFS) for radar avoidance — APs can start transmitting immediately without radar detection delays.
- **Higher Throughput in Practice:** The combination of wide channels (160 MHz), clean spectrum, and OFDMA means real-world throughput in the 6 GHz band is significantly higher than what is achievable in congested 5 GHz deployments.
- **Ideal for High-Density Deployments:** Stadiums, airports, convention centers, and office buildings benefit enormously from the additional capacity and clean spectrum of 6 GHz.

Limitations of 6 GHz:

- **Shorter range:** Higher frequencies have less wall penetration and shorter range compared to 2.4 GHz and 5 GHz.
- **Device compatibility:** Requires new Wi-Fi 6E-certified devices — older devices cannot connect in the 6 GHz band.
- **Regulatory variation:** The full 1200 MHz of 6 GHz spectrum is not available in all countries — availability depends on regional regulatory approval.

5. Compare and contrast Wi-Fi 6 and Wi-Fi 6E in terms of range, bandwidth, and interference.

Answer:

Aspect	Wi-Fi 6 (802.11ax)	Wi-Fi 6E (802.11ax + 6 GHz)
Frequency Bands	2.4 GHz and 5 GHz	2.4 GHz, 5 GHz, and 6 GHz
Standard	802.11ax	802.11ax (same standard, extended band)
Available Spectrum	~570 MHz total (2.4 + 5 GHz)	~1770 MHz total (adds 1200 MHz at 6 GHz)
Max Channel Width	160 MHz	160 MHz (up to 320 MHz in Wi-Fi 7)
Non-overlapping 160 MHz channels	2 (in 5 GHz)	7 (in 6 GHz alone)
Range (2.4 GHz)	Long range, good wall penetration	Same — 2.4 GHz still available
Range (5 GHz)	Medium range	Same — 5 GHz still available
Range (6 GHz)	Not applicable	Shorter than 5 GHz — higher frequency attenuates more through walls
Legacy Device Support	2.4/5 GHz: Yes (802.11b/g/n/ac devices)	6 GHz: No — only Wi-Fi 6E devices allowed
Interference (2.4 GHz)	High — Bluetooth, microwaves, legacy Wi-Fi	Same — 2.4 GHz still congested
Interference (5 GHz)	Moderate — some DFS, legacy devices	Same — 5 GHz still has legacy devices
Interference (6 GHz)	Not applicable	Minimal — no legacy devices, fresh spectrum
DFS Required	Yes (for some 5 GHz channels)	No (in most regions for 6 GHz)
Throughput in Practice	Limited by congestion in 2.4/5 GHz	Higher — clean 6 GHz spectrum enables full OFDMA efficiency
Device Requirement	Most Wi-Fi 6 devices	Requires Wi-Fi 6E certified device with 6 GHz radio
Best Use Case	General home/office use, backward compatibility needed	High-density venues, low-latency applications, future-proof networks

Key Takeaway:

Wi-Fi 6 and Wi-Fi 6E use the same 802.11ax standard and the same technologies (OFDMA, TWT, BSS Color, MU-MIMO). The difference is purely the addition of the 6 GHz band in Wi-Fi 6E. The 6 GHz band offers more bandwidth and far less interference, but at the cost of reduced range and requiring newer hardware. For environments where maximum performance is needed (enterprise, high-density), Wi-Fi 6E is clearly superior. For general use where backward compatibility and range matter, Wi-Fi 6 without 6 GHz may be more practical.

6. What are the major innovations introduced in Wi-Fi 7 (802.11be)?

Answer:

Wi-Fi 7 (802.11be), branded as Wi-Fi 7, is the latest Wi-Fi generation ratified in 2024. It builds on Wi-Fi 6/6E with several groundbreaking innovations aimed at achieving extremely high throughput (EHT) and ultra-low latency.

1. Multi-Link Operation (MLO):

- The most significant innovation in Wi-Fi 7. MLO allows a device to simultaneously establish links on multiple bands and channels (e.g., 5 GHz and 6 GHz at the same time).
- Traffic can be split across multiple links for higher aggregate throughput, or a link can be used as a backup for lower latency and higher reliability.
- Fundamentally changes Wi-Fi from a single-link to a multi-link architecture.

2. 320 MHz Channel Width:

- Wi-Fi 7 supports channels up to 320 MHz wide in the 6 GHz band — double the 160 MHz maximum of Wi-Fi 6/6E.
- A 320 MHz channel provides roughly 2× the throughput of a 160 MHz channel with the same modulation.
- Only possible in the 6 GHz band due to the large available spectrum.

3. 4096-QAM (4K-QAM):

- Wi-Fi 7 increases modulation from 1024-QAM (Wi-Fi 6) to 4096-QAM.
- 4096-QAM encodes 12 bits per symbol vs 10 bits for 1024-QAM — a ~20% increase in peak data rate.
- Requires excellent signal quality (very high SNR) — works best at close range.

4. Multi-RU (Multiple Resource Units):

- In Wi-Fi 6, each client could only be assigned one RU per transmission.
- Wi-Fi 7 allows a single client to be assigned multiple non-contiguous RUs in the same OFDMA transmission — increasing flexibility and throughput for individual users.

5. Preamble Puncturing:

- Allows the AP to use a wide channel (e.g., 320 MHz) but puncture (skip) specific 20 MHz sub-channels that are occupied by interference or radar.
- In Wi-Fi 6/6E, if any part of a wide channel is busy, the entire wide channel had to be abandoned.
- Puncturing dramatically improves spectrum utilization in real-world environments.

6. 16 Spatial Streams (MU-MIMO):

- Wi-Fi 7 supports up to 16 spatial streams (vs 8 in Wi-Fi 6/6E) — allowing more simultaneous MU-MIMO users or higher per-client throughput.

7. Enhanced Link Adaptation and Retransmission:

- Improved mechanisms for adapting MCS per-link in real time and for retransmitting only failed sub-channels in MLO environments.

Performance Summary:

Innovation	Wi-Fi 6/6E Maximum	Wi-Fi 7 Maximum
Max Throughput	9.6 Gbps	46 Gbps
Channel Width	160 MHz	320 MHz
Modulation	1024-QAM (10 bits/symbol)	4096-QAM (12 bits/symbol)
Spatial Streams	8	16
Multi-Link	Not supported	MLO — simultaneous multi-band
RU per client	Single RU only	Multiple non-contiguous RUs
Puncturing	Not supported	Supported

7. Explain the concept of Multi-Link Operation (MLO) and its impact on throughput and latency.

Answer:

What is Multi-Link Operation (MLO)?

Multi-Link Operation (MLO) is the defining feature of Wi-Fi 7 (802.11be). It allows a Wi-Fi device (both the AP and client) to establish and maintain multiple simultaneous wireless links across different frequency bands and/or channels at the same time — treating them as a single logical connection.

Before MLO, a Wi-Fi device could only be associated on one band at a time (even on a tri-band AP). Band steering would move the device between bands, but only one link was active for data. MLO changes this fundamentally.

How MLO Works:

- An MLO-capable AP and client negotiate multiple links during the association process — for example, one link on 5 GHz and another on 6 GHz.
- Each link has its own physical channel, frequency, and PHY parameters.
- The MAC layer is shared (single MAC address, single Block ACK session, single sequence number space) but traffic is distributed across multiple PHY links.
- Three MLO modes exist: Simultaneous Transmit and Receive (STR) — transmit and receive on different links at the same time; Enhanced Multi-Link Single Radio (EMLSR) — single radio switches between links dynamically; Enhanced Multi-Link Multi-Radio (EMLMR) — multiple radios active simultaneously.

Impact on Throughput:

- Aggregated Bandwidth: Traffic is split across multiple links, so the effective throughput is the sum of all active link capacities. A device using 5 GHz (2.4 Gbps) + 6 GHz (4.8 Gbps) simultaneously achieves ~7.2 Gbps aggregate.
- Load Balancing: The MAC scheduler distributes frames across links based on channel conditions — if one link is congested, more traffic shifts to the other.
- Higher Peak Rates: MLO combined with 320 MHz channels and 4096-QAM enables the 46 Gbps theoretical maximum of Wi-Fi 7.

Impact on Latency:

- Latency Reduction via Redundancy: Time-sensitive frames (VoIP, gaming, AR/VR) can be sent on whichever link has the least congestion at that moment — avoiding queuing delays.
- Link Redundancy: If one link experiences interference or a temporary fade, traffic immediately moves to another link without reconnection delays.

- No Roaming Required: Before MLO, moving between bands required re-association (even with 802.11r, there is some delay). MLO eliminates this — all bands are always active.
- Target: Sub-1 ms latency for real-time applications — a major improvement over Wi-Fi 6's typical 5–10 ms latency in loaded networks.

8. What is the purpose of 802.11k and v, and how does it aid in roaming?

Answer:

802.11k and 802.11v are two amendments to the 802.11 standard that provide infrastructure-assisted roaming — enabling clients to make smarter, faster roaming decisions with help from the network. Together with 802.11r, they form the three pillars of seamless enterprise roaming.

802.11k – Radio Resource Management (RRM):

- Purpose: Gives clients visibility into the radio environment so they can identify better APs to roam to, before their current connection deteriorates.
- Neighbor Report: The client sends a Neighbor Report Request to its current AP. The AP responds with a Neighbor Report — a list of nearby APs (their BSSIDs, channels, and capabilities) that the client should consider for roaming.
- Without 802.11k: The client must scan all channels to find neighboring APs — this takes time and interrupts data flow.
- With 802.11k: The client already knows which channels to scan and which BSSIDs to target, making its scan targeted and fast.
- Beacon Report: The AP can also request that the client measure and report the signal strength of nearby APs — helping the network build a radio map.
- Benefit for Roaming: Reduces the time the client spends scanning by giving it a pre-built list of candidate APs. The client can begin evaluating roaming candidates much earlier.

802.11v – BSS Transition Management:

- Purpose: Allows the AP (network) to actively guide or instruct the client on when and where to roam — giving the network control over client roaming decisions.
- BSS Transition Management Request: The AP sends this frame to a client to suggest (or in some cases strongly request) that it roam to a specific AP. This is sent when: The client's signal is weak, the current AP is overloaded, or network load balancing is needed.
- The request includes a list of candidate APs with their BSSIDs and signal quality information.
- Disassociation Imminent: The AP can set a flag indicating it will disassociate the client soon — forcing the client to prepare for roaming immediately.
- Benefit for Roaming: The network can proactively move clients before they degrade to a poor signal state. Without 802.11v, clients often 'sticky' — staying connected to a distant AP with a poor signal rather than roaming to a closer one.
- Solves the Sticky Client Problem: The most practical benefit of 802.11v in enterprise deployments.

How 802.11k and v Work Together for Roaming:

- 802.11k provides the client with a neighbor list so it knows where to roam.
- 802.11v allows the AP to tell the client when to roam and suggest the best target AP.
- Combined, the client has both the knowledge (k) and the direction (v) to make fast, intelligent roaming decisions without wasting time scanning unknown channels.

9. Explain the concept of Fast BSS Transition (802.11r) and its benefit in mobile environments.

Answer:

What is 802.11r – Fast BSS Transition (FT)?

802.11r (Fast BSS Transition) is an IEEE 802.11 amendment that enables a wireless client to roam from one AP to another within the same ESS (Extended Service Set / same SSID) with significantly reduced roaming time — particularly by pre-establishing security keys with the target AP before the actual roam occurs.

The Problem Without 802.11r:

In standard WPA2 roaming, when a client moves to a new AP, it must complete a full re-authentication sequence: 802.11 Authentication → Association → Full EAPOL 4-Way Handshake (and for 802.1X, a complete EAP exchange with the RADIUS server). This can take 500 ms to several seconds — causing dropped VoIP calls, video stutter, and session interruptions.

How 802.11r Works:

- 802.11r introduces a key hierarchy that allows cryptographic keys to be pre-distributed to neighboring APs via the wired network before the client roams.
- R0KH (R0 Key Holder): Typically the WLC or a designated AP — holds the master roaming key (PMKR0).
- R1KH (R1 Key Holder): Each AP — holds a per-client, per-AP key (PMKR1) pre-derived from the PMKR0.
- When a client decides to roam, it sends an FT Authentication Request to the target AP. Because the target AP already has the PMKR1 for this client (pre-distributed via the R0KH), it can immediately complete authentication and key installation in just 2 frames — no RADIUS server round-trip needed.
- FT Over-the-Air (FT-OTA): The client communicates directly with the target AP for fast transition.
- FT Over-the-DS (FT-OTD): The client communicates with the target AP via the current AP (through the distribution system) before physically moving.

Benefits in Mobile Environments:

- Roaming Time Reduction: Standard roaming can take 500 ms–2 s. With 802.11r, roaming completes in under 50 ms — often as low as 10–20 ms.
- VoIP and Real-Time Applications: Voice over Wi-Fi (VoWi-Fi), video conferencing, and real-time control applications require roaming time under 50 ms to avoid call drops or audio gaps. 802.11r makes this achievable.
- No RADIUS Server Delay: The key pre-distribution means the target AP does not need to contact the RADIUS server during roaming — the biggest source of roaming delay in WPA2-Enterprise.
- Seamless Mobility: Users moving through a building (hospital nurses, warehouse workers, students) experience no perceptible service interruption.
- Foundation for Wireless Voice: 802.11r is a requirement for Wi-Fi Calling (VoWi-Fi) certification and is listed as mandatory in Cisco's voice-grade wireless designs.

10. How do 802.11k/v/r work together to provide seamless roaming in enterprise networks?

Answer:

In enterprise networks, 802.11k, 802.11v, and 802.11r are designed to work together as a complete roaming framework — often referred to as the '802.11kvr' trio or 'Fast Transition with BSS Transition Management'. Each standard addresses a different phase of the roaming process.

The Three Phases of Roaming and How Each Standard Helps:

Phase	Standard	Role
1. Discovery – Finding where to roam	802.11k	AP provides a Neighbor Report so client knows which channels and

Phase	Standard	Role
		BSSIDs to scan — eliminates full channel scan delay
2. Decision – Deciding when to roam	802.11v	AP sends BSS Transition Management Request to suggest or force the client to roam to a specific AP at the right time
3. Execution – Performing the roam	802.11r	Fast key exchange with the target AP using pre-distributed keys — roaming completes in <50 ms without full re-authentication

End-to-End Roaming Sequence with 802.11kvr:

- Step 1 (802.11k): The client associates to AP1. At some point, it requests a Neighbor Report from AP1 and receives a list of nearby APs (AP2, AP3...) with their channels and BSSIDs.
- Step 2 (Background Scan): Using the neighbor list, the client performs a targeted scan only on the channels where AP2 and AP3 operate — much faster than scanning all channels.
- Step 3 (802.11v): As the client moves away from AP1 and RSSI drops to a threshold, AP1 sends a BSS Transition Management Request suggesting AP2 as the best target, based on its knowledge of client signal levels and AP load.
- Step 4 (802.11r): The client sends an FT Authentication Request to AP2. AP2 already has the pre-derived PMKR1 key for this client (shared by the R0KH). The client completes authentication and key installation in 2 frames.
- Step 5 (Roam Complete): The client is now associated to AP2 with encryption active. The entire roam — from decision to full connectivity — takes under 50 ms. The VoIP call or video stream continues without interruption.

Why All Three Are Needed:

- 802.11k alone: The client knows where to roam but still roams too slowly (full 4-way handshake) and may roam at the wrong time.
- 802.11v alone: The network can suggest roaming targets but the client still wastes time scanning and performing full re-authentication.
- 802.11r alone: Roaming is fast when it happens, but the client may not know when to roam or where the best AP is — leading to sticky client behavior.
- All three together: The client gets the neighbor list (k), gets told when and where to roam (v), and executes the roam in under 50 ms (r). This is the complete solution for seamless enterprise mobility.

Enterprise Deployment Note:

In Cisco enterprise deployments, 802.11k/v/r are enabled together on the WLAN (SSID) configuration. The WLC acts as the R0KH (master key holder) and pre-distributes roaming keys to all APs (R1KHs) in the mobility domain. This ensures that regardless of which AP the client roams to, fast transition is available. 802.11k/v/r support is also required for Wi-Fi CERTIFIED Voice certification and is strongly recommended for any network supporting mobile devices.